A3-PS029-TC400

Application-Oriented Network Quality Evaluation Using QoO

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Problem Description

• A fast network is not always a good network: gaming and video streaming can experience the same connection very differently.

• Average speed and delay hide jitter and bursty packet loss.

• Quality of Outcome (QoO) scores a network against an application profile, including latency, loss and throughput requirements.

• Key questions: How does profile choice change the verdict? Can QoO distinguish steady from jittery delay and random from bursty loss? Are results reproducible and trustworthy?

Background

• Latency is delivery time; jitter is variation in delay; packet loss is data that never arrives; throughput is data transferred per second.

• Gaming needs small, timely updates. Streaming needs higher throughput and can buffer through small delays.

• QoO is an IETF IPPM Internet-Draft, building on standard one-way delay and loss metrics.

• Delay variation and loss patterns matter beyond averages.

• We reuse the supplied Docker QoO testbed and Linux NetEm to emulate delay, jitter and loss, then compare QoO verdicts with measured application behavior.

Literature & RFC Review

- QoO Internet-Draft: application-profile scoring framework; used unchanged.
- RFC 7679 / RFC 7680: standard one-way delay and loss metrics.
- RFC 3393 / RFC 3357: delay variation and loss-pattern metrics.
- RFC 6349: TCP throughput testing; informs the iperf3 cross-check.
- Linux NetEm provides controlled impairments; metamorphic testing checks whether outputs change sensibly as inputs change.
- Research gap: audit how consistently QoO behaves across profiles and bursty network conditions.

Proposed Solution

A reproducible harness scores the same emulated networks under two application profiles and audits the result.

TWO APPLICATION PROFILES

A ONLINE GAMING

Needs: low, steady delay
Latency: p50 ≤30 ms; p95 ≤60 ms; p99 ≤100 ms
Loss ~1% · Throughput ~1 Mbps

B ADAPTIVE STREAMING

Needs: high throughput
Latency: p50 ≤200 ms; p95 ≤500 ms; p99 ≤1 s
Loss ~1% · Throughput ~4–5 Mbps

SIX SCENARIOS (+ CLEAN REFERENCE)

ID	IMPAIRMENT	PURPOSE
S1	+20 ms fixed delay	Small delay
S2	+20 ms avg, ±15 ms jitter	Pair with S1
S3	+100 ms fixed delay	Gaming may fail
S4	1% random loss	Pair with S5
S5	1% bursty loss	Pattern differs
S6	+30 ms, ±10 ms jitter, 1% loss	Poor link

App metrics: gaming—UDP deadline-miss rate · streaming—goDASH rebuffering and bitrate switches.

METHOD & OUTPUTS

• Run both profiles back-to-back 5–10 times per scenario; sample latency ≥10 Hz for ≥60 s.
• Record QoO, latency, throughput, app metrics and JSON metadata (protocol, endpoints, timestamps, duration, settings, versions). Output: dataset, dashboard, report and profile comparison. Keep QoO unchanged; report measurable results only.

Optimization Proposed

Extensions improve confidence in QoO scores, explain their meaning, and make runs repeatable.

AREA	BASELINE (QoO as supplied)	WITH OUR EXTENSIONS
Trust in score	Taken as given	QoO Audit: sanity/metamorphic tests, ping/iperf3 cross-check, 95% confidence intervals, disagreement table, hypotheses and rank correlation.
Meaning of score	One number per run	Insight Lab: agreement map showing where QoO and application metrics agree.
Reuse	Manual runs	Proof Kit: one-command run (make run-all), standard results format and “how to read the scores” guide.

HYPOTHESIS, EXTENSIONS & FUTURE WORK

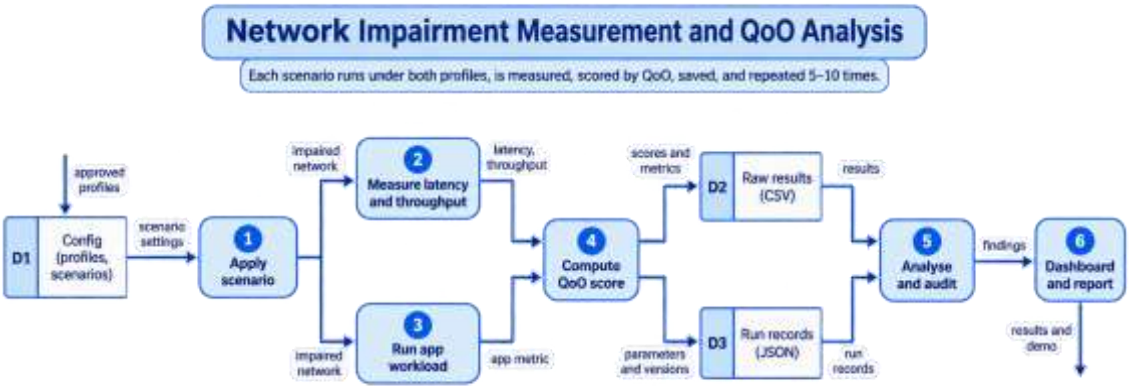
- Hypothesis: if QoO uses only average loss, S4 and S5 may score similarly even when gaming outcomes differ.
- If time allows: remaining tests, a small fuzzer and a network-weather run.
- Future work: smallest-fix finder, real traffic, trace replay, threshold sensitivity and plug-in audit kit.
- Scope: label extras “Extended study”; keep them separate from official scenarios and get mentor approval.

Solution Architecture & Design

Config → Harness → Docker testbed (NetEm + QoO tool) → Storage (JSON/CSV) → Audit → Dashboard



Data Flow Diagram



Deployment Timeline

- 14–15 Oct — Submit proposal; testbed running; verify burst loss and jitter; mentor approves profiles; commit hypotheses and results format.
- 16–19 Oct — Build harness and run records; run all 12 profile/scenario combinations once; create gaming workload and first dashboard.
- 20–22 Oct — Complete repeats; ping/iperf3 cross-check; verify matched pairs.
- 23–25 Oct — Run audit tests; prepare disagreement table, rank correlation and agreement map.
- 26 Oct — Finish report, README, score-reading guide, clean-machine run and backup demo video.
- 27 Oct — Final commit and rehearsals; 28–31 Oct reserved for fixes.
- Roles: Bhavya — testbed, scenarios, harness, workloads, GitHub. Kissanthini — analysis, tests, dashboard, report, slides and demo. Complete required work before audit and extras.

References

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6. S. Hemminger, Network Emulation with NetEm, linux.conf.au, 2005.
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Thank You!

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